Exhibition - Blind Spots
Identifying Known and Unknown Biases

We propose to develop an immersive and interactive exhibition that takes research findings into the public domain. Exhibitions help to generate press around a topic that might go unnoticed in the community, showcase the work behind a research project, while at the same time getting on the radar of those making the policy decisions that affect us all.

Designed to travel to multiple locations and be seen by diverse people, this exhibition creates a more tactical (touching/ moving physical objects) experience, humanizing the problem of biases in Artificial Intelligence (AI) algorithms. Systemic social biases, often unnoticed yet deeply embedded in society, unevenly favor certain groups based on individual traits. These biases are not just theoretical concerns; they are increasingly embedded in these systems, influencing everything from healthcare and politics to justice systems and consumer behavior. The omnipresence of these decision-making systems calls for civic engagement, and for that **we propose a three-part exhibition: two of those parts in a closed location and a third that should take place outdoors**.

1. **Share the knowns:** This initial part of the exhibition addresses the first two foundational questions of our project: “Can we systematically identify biases in datasets and machine learning algorithms (MLAs)?” and “Can we minimize the negative social impacts of these biases while maintaining the advantages of MLAs?” Here, we invite visitors to explore the current state-of-the-art methods for identifying and reducing social biases. This exploration is facilitated through a variety of mediums, including text, videos, images, and infographics.

   At the center of this section is an interactive experience based on FARE_Audit and an existing web-crawler system. Visitors begin by creating their own avatars, choosing options such as gender (male, female, other), age, and answering some general questions. This process is tailored to help visitors understand how different demographic groups might encounter varying biases in the digital world. [Visitors are thoroughly informed about how their data is used and their rights concerning it, including access and erasure, ensuring a fully GDPR-compliant experience.] After creating their avatars, participants engage with the exhibition by entering search queries into an interface. The web-crawler, programmed to mimic the browsing behavior of diverse user profiles, then retrieves data from the internet based on each avatar's unique characteristics. The results, reflective of each avatar's profile, give visitors an experience of how search outcomes and digital information access can significantly vary with personal demographics, thereby highlighting the presence of digital biases.

   A key feature of this segment is its ability to perform comparative analyses. Using real-time data visualizations, the exhibition contrasts the search experiences of individual avatars against other profiles in the system or against an aggregate of other users. This direct comparison not only underscores the disparities in information access but also offers a tangible representation of biases that could be perpetuated by AI systems. This demonstration extends into an explanation of how trained machine learning models might amplify these existing biases, providing a comprehensive view of the cycle of bias within AI systems.

1. **Speculating on the Unknowns**: the second segment of the exhibition, tries to address the third and fourth main questions of this project from a contemplative space: can we uncover new biases and learn about society in the process? And, how should algorithmic bias be minimized, or what does it mean to have fair AI? This part of the exhibition is not merely a display; it creates reflective spaces that challenge visitors to think beyond the present, to a society intricately entwined with AI—ranging from utopian visions of a bias-free society to dystopian realities where biases are deeply institutionalized and normalized.

   To lay the groundwork for this introspective part of the exhibition, several workshops will be conducted that bring together a diverse group of stakeholders, including AI researchers, developers, ethicists, sociologists, community leaders, advocates, and psychologists. The dialogue generated in these workshops, combined with the research already conducted, will inform, and inspire a selected group of young Portuguese writers. These emerging talents will be given the opportunity to construct provocative speculative scenarios, designed not to dictate but to inspire the visitors to engage in conversations about potential futures.

   Using visual storytelling, the essays produced by the writers will be narrated while visual clues of those imaginary scenarios will be displayed at the same time. This technique will be used to resonate with the people at an emotional level, drawing out feelings and reflections rather than technical contemplation. This part emphasizes the complexities and uncertainties in envisioning a future where AI can either be a tool for social betterment or for deepening existing divides. These speculative scenarios aim to instill in every participant the recognition that addressing these biases is not just a technological challenge, but a civic duty and a collective choice.

2. **Democratizing the imagination of our future:** In response to William Gibson's observation that "The future is already here, it's just not evenly distributed," this segment of the exhibition extends beyond the confines of a closed space into the fabric of the city. Interactive and analogue data devices, conceived for outdoor display, will serve a dual purpose: they will capture public opinion through innovative voting mechanisms and increase awareness of social biases and the implications of AI.

   These devices, strategically situated in various locations around the city, are designed to encourage public engagement and gather real-time feedback. Crafted by the experienced Colombian designer Jose Duarte, well known for his "debate detonators," these mobile installations are meant to spark public discourse and directly involve individuals in the conversation. Their straightforward, portable design and bold visual appeal aim to engage a broad audience, reaching those who might not typically visit an exhibition space, and seamlessly integrating into the daily routes of city-dwellers.

   Owing to the participatory nature of these pieces, this initiative will commence in Lisbon, Portugal, as a prototype that, if successful, could be replicated and tailored to the unique contexts of other cities.

In conclusion, the exhibition is designed to anticipate and respond to technological transformations in society. At the end of the first two parts, visitors are encouraged to share their insights through an online survey accessed via QR code, and real-time feedback boards will be available for immediate community engagement. This feedback will be instrumental in continually updating and improving the quality of the exhibition.

In essence, this exhibition is a dynamic and collaborative examination of AI's societal impact, aiming to educate, provoke thought, and inspire action towards a more equitable technological future.